

Muhammad Taha Khan

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Professional Summary

Backend Engineer with **1+ years of experience** designing and optimizing **high-performance distributed systems, robust infrastructure**, and **agentic AI workflows**. Proven track record of **leading engineering teams (3 to 4 members)** to ship **enterprise-grade client projects** from inception to production. Expert in **asynchronous messaging, cryptographic security**, and **advanced RAG architectures**. BS Computer Science, NUCES-FAST Karachi (June 2026, CGPA 3.76).

Core Competencies

- **Engineering Leadership:** Proven track record of leading small teams to deliver enterprise-grade software.
- **System Architecture:** Expertise in distributed systems, robust infrastructure, and intelligent workflows.
- **Cross-Functional Delivery:** Seamlessly bridge backend, DevOps, and AI domains to solve complex business problems.

Technical Skills

- **Languages & Runtimes:** Node.js (Express, Fastify), JavaScript/TypeScript, Python, React, Next.js
- **Databases & Storage:** PostgreSQL (incl. pgvector), Redis, Neo4j, Apache AGE
- **Frameworks & AI:** gRPC, LangGraph, CrewAI, RAG pipelines, Knowledge Graphs, Claude Code (MCP, Plugins, Agents, Structured Collaboration)
- **Infra & DevOps:** Docker, AWS, Terraform, Kubernetes, PM2
- **Messaging & Tools:** RabbitMQ, Kafka, Firebase, Google OAuth, Nodemailer

Work Experience

- **Associate Backend Developer at 360XpertsSolutions** **June 2025 to Present**
 - Advanced from backend trainee to Associate Developer within 2 months; led teams of 3 to 4 engineers to successfully ship **6+ production-grade client projects** ahead of schedule.
 - **Core Infrastructure & Architecture:** Architected a highly scalable, fully layered boilerplate framework enforcing Separation of Concerns (SoC) and Inversion of Control (IoC) via advanced Dependency Injection (DI). Implemented strict data serialization validation, robust database layers, and **event sourcing**, orchestrating **RabbitMQ, Kafka**, and **Redis**; accelerated team-wide feature bootstrapping velocity by **70%**.
 - **Secure Chat Architecture:** Architected a multi-tenant messaging backend utilizing hybrid **AES-256 symmetric** and **NaCl/Elliptic Curve asymmetric encryption** with per-message key rotation, ensuring zero-knowledge data-at-rest protection while maintaining **real-time messaging with <500ms latency**.
 - **AI-Assisted E-Learning Platform:** Developed a scalable, multi-tenant educational platform featuring robust **RBAC** and **event sourcing**, utilizing **PostgreSQL, RabbitMQ**, and **Redis**. Integrated advanced **RAG** pipelines for dynamic, AI-driven course generation and personalized learner experiences, successfully deploying **100+ automated courses** and driving administrative audit analyses with **85% accuracy**.
 - **Intelligence & RAG Layers:** Architected the intelligence layer for chat systems using **CrewAI** and **LangGraph** state maps; engineered advanced memory layers and context management pipelines for accurate knowledge representation, achieving **70% query and context extraction accuracy**.
 - Mentored junior trainees on system design patterns, distributed debugging, and CI/CD pipelines to ensure team-wide alignment with production-level code standards.

Education

- **BS-CS (June 2026):** NUCES-FAST Karachi **Graduated; CGPA 3.76/4.0**
 - **FYP:** Addressing Challenges of Person Re-Identification: Proposed a **light weight Person Re-identification pipeline** tackling **occlusion, pose variation**, and **illumination variation**, a gap unaddressed in existing literature of a highly researched domain.